How did trade liberalisation impact economic growth in the period 1988-2023?

Radost Trakiyska

There has been a wide range of theories and beliefs that have shaped the views on how countries should enforce trade policy. In early modern Europe, sovereigns would pursue mercantilism, a form of economic system aimed at maximising exports and minimising imports in order to accumulate wealth. Mercantilist policies were linked to imperial expansion, as colonial trade was controlled to serve the interests of the colonial power, often through restrictive trade channels and unfavourable terms for the colonies. Eli Heckscher argued that mercantilism primarily enhanced the power and wealth of the nation state, but did not ultimately improve the welfare of its people. Heckscher viewed trade as a zero-sum game, thus encouraging protectionism and high tariff policies. This view stands in contrast to later classical economists, most notably Adam Smith, who rejected mercantilist reasoning and advocated for free trade.

Adam Smith was among the first to propose that international trade leads to economic growth. He argues that specialisation and division of labor increase productivity, thereby making societies wealthier. On this basis, Smith advocates in favour of free trade as a mechanism through which countries could achieve higher incomes, and ultimately, economic convergence. David Ricardo later developed his theory of comparative advantage, which supported free trade by showing that countries benefit from specialising in goods they can produce at lower opportunity costs compared to other goods they can produce, and importing other goods. This implies that trade protectionism is never beneficial to a country's economy as a whole, as it forces allocation of resources towards less efficient uses by producing goods that are not in its comparative advantage. These classical thinkers laid the theoretical foundation of modern analysis of the relationship between trade and growth.

Investigating the relationship between trade barriers and economic growth is complicated due to the nature of operationalising the variable trade restriction. There have been concerns with methods of tariff and nontariff data measurement, aggregation, endogeneity, quality (Rodriguez & Rodrick, 2000; Teti, 2024), and several authors have turned to alternative empirical methodologies to test the impact of trade on growth. A landmark paper displaying this is 'Economic reform and the process of global integration' by Jeffrey Sachs & Andrew Warner (1995).

The largest study investigating this topic was done by Sachs & Warner (1995), investigating the topic of economic convergence in 1970-1989, looking at a sample of 111 countries. Their paper aims to assess the impact of global integration on economic growth in developing and developed countries, and attempt to address economic convergence through it. Workhorse economic models predict that poor countries should grow faster than rich countries, due to poorer countries' ability to import modern technology and capital from richer countries, and catch up over time. Therefore, it is expected to see an economic convergence in the world economy; in other words, the income gap between countries should close as time goes on. However, this is not observed in reality. Sachs & Warner (1995) put forward the hypothesis that this is explained by trade openness - open economies converge, whereas closed economies do not. Poorer countries have historically been closed to the world, and Sachs & Warner (1995) theorise that this fact would explain why a lack of convergence is observed.

They aim to test whether economies that trade liberalise would tend to converge with the global economy.

The methodology that Sachs & Warner (1995) utilised is the following. Their analysis hinges on the system they've defined to categorise countries' economies as open or closed, constructing a dummy variable to indicate whether a country has an open or closed trade policy. They provide several criteria, which characterise major types of restrictions on trade, and they classify the timing of an economy's trade liberalisation and its effect on subsequent growth. A country that meets at least one of the following characteristics would be categorised as having a closed trade policy:

1. Average tariff rates of 40 percent or more.
2. Nontariff barriers covering 40 percent or more of trade.
3. A black market exchange rate that is depreciated by 20 percent or more relative to the official exchange rate, on average, during the 1970s or 1980s.
4. A socialist economic system.
5. A state monopoly on major exports.

Sachs & Warner (1995) find a significant relationship between trade openness and economic growth, both for developing and developed countries. They show that open economies grew significantly faster than closed ones in 1970-1989, and developing countries grew faster than developed countries. They conclude that economic convergence occurs only among countries integrated into the global economy.

Sachs & Warner's (1995) model faced criticism in a paper published by Rodriguez & Rodrick (2000). From the variables Sachs & Warner (1995) used for categorising a country as closed, the black market premium (BMP) and state-monopoly (MON) variables tended to largely determine the value of the openness dummy, and therefore the categorisation of the country. Black market premium measures the degree of rationing in the foreign currency market - a high BMP indicates that foreign currency trades at a much higher price on the black market than at the official exchange rate. This means that firms cannot freely access foreign currency at the official exchange rate, raising the cost of imports and making this variable a form of import control. The state monopoly on major exports criteria was included for cases when governments tax major exports, leading to a reduced level of both exports and imports.

Rodriguez & Rodrick (2000) reproduced Sachs & Warner's (1995) regression, and instead of the openness dummy, they regressed each variable making up the dummy separately instead of it. They found that BMP and MON are individually significant in predicting economic growth, however, the remaining three variables are not (Rodriguez & Rodrick, 2000). They proceeded to test the difference between the original dummy and a reconstructed openness dummy variable containing only the BMP and MON criteria, finding that the model proved significant, whereas a model with the remaining three variables instead of the dummy was not (Rodriguez & Rodrick, 2000). In other words, the black market premium and state-monopoly variables dominate the value of the dummy variable in comparison to tariff rates, nontariff barriers, and socialist/non-socialist classifications. Furthermore, the model with only BMP and MON differed in only 6 cases when compared to the model with the original openness dummy, as opposed to the 31 cases between the model with the remaining three variables and the original (Rodriguez & Rodrick, 2000).

In 2008, Warcziarg & Welch (2008) published a follow-up to Sachs & Warner's research, replicating their analysis using the same data source with a larger sample of 141 countries, looking at the period from 1950-1998. The specification of the model is the same as the original research. Warcziarg & Welch (2008) found the same significant results as Sachs & Warner (1995) for the extended time period, finding that countries that liberalised their trade policies saw a 1.5% higher average growth rate per year, compared to before liberalisation. The country status for the newly extended 1990-1999 decade faces the same shortcomings as pointed out by Rodriguez & Rodrick (2000) in the original study. To illustrate this, a model with a dummy variable containing only BMP and MON differed in only 2 cases compared to the model with the dummy variable containing all criteria, and a model using the three remaining variables - tariff rate, nontariff barriers, and socialist economic system - resulted in 38 differing cases compared to the model containing all criteria (Warcziarg & Welch, 2008).

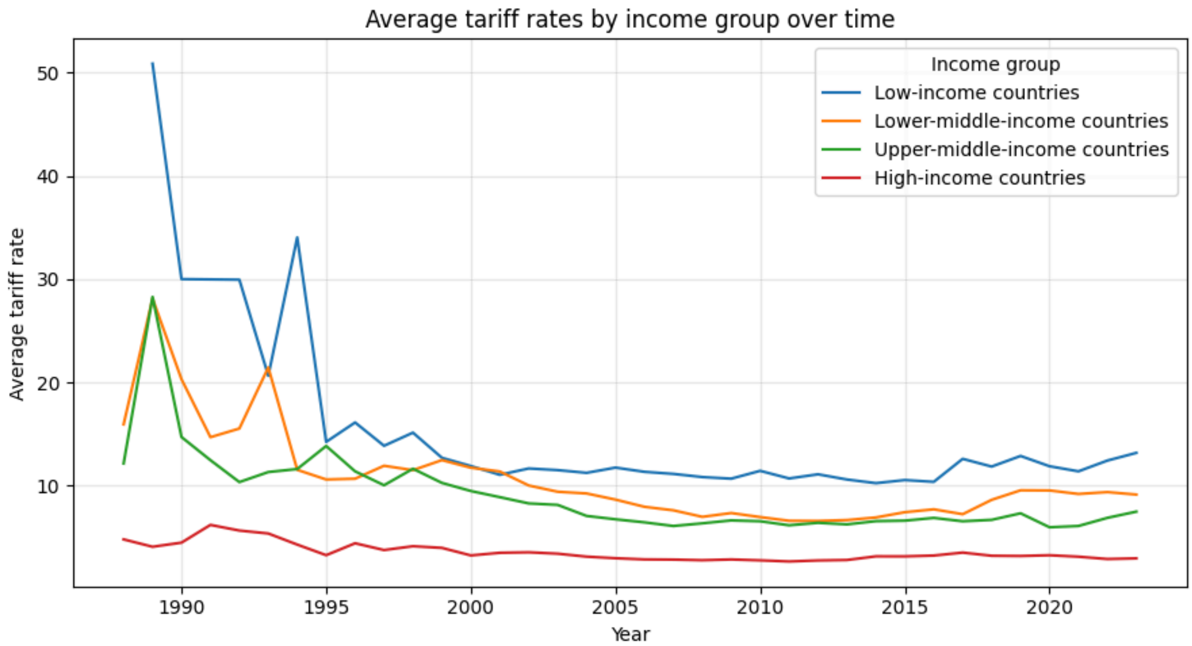
Due to Rodriguez & Rodrick's (2000) critiques to the original study, this analysis aims to replicate the Sachs & Warner (1995) and Wacziarg & Welch's (2008) model with several notable differences. Firstly, it extends the analysis by 25 years by examining the period 1988-2023. Secondly, it aims to investigate whether a model where the trade openness dummy is replaced by tariff rates would still prove significant. This is because, from the five criteria set by Sachs & Warner (1995), tariff rates had a very small impact on the openness dummy variable, despite being the most direct measure of trade policy (Rodriguez & Rodrick, 2000). This analysis avoids a binary classification of countries' liberalisation status and rather examines the direct relationship between tariff rates and economic growth.

Historical Background

In the late 1980s and 1990s, developing countries have been increasingly globalising by opening up their economies to trade and investment. Figure 1 shows the drastic reduction in average tariff rates in low-income countries throughout the 1990s, going from a peak of over 50% to settling at a rate around 10%. Lower-middle-income countries and upper-middle-income countries are observed to also experience a decrease in their average tariff rates from 1988-1996, at first varying around 15% before also diminishing to a constant rate of around 10%. High-income countries' tariff rates stay largely constant at around 3% throughout the whole period.

Figure 1

Line plot showing trend of average tariff rates by income group in the period 1988-2023



The wave of liberalisation that occurred in the 1980s and 1990s was due to policy changes for keeping a balance of payments. Many of the trade restrictions that kept economies closed until that period were initially imposed to manage their payments balance under fixed exchange rates (Irwin, 2025). During the 1930s Great Depression, governments limited spending on imports to avoid using up the country's foreign exchange reserves instead of devaluing their own currency to adjust the exchange rate (Irwin, 2025). In the mid-1980s, foreign exchange shortages worsened and made this strategy unsustainable, leading governments to abandon trade restrictions and instead focus on exchange rate adjustments that promoted exports and limited imports (Irwin, 2025). The shift to using devaluation to balance payments instead of trade policy was the reason for the large economic trade liberalisation in these two decades.

Consequently, Figure 2 shows corresponding large increases in countries' trade-to-GDP ratios. The general trends show an increase in exports and imports after the late 1990s, with high-income countries naturally seeing the largest spike.

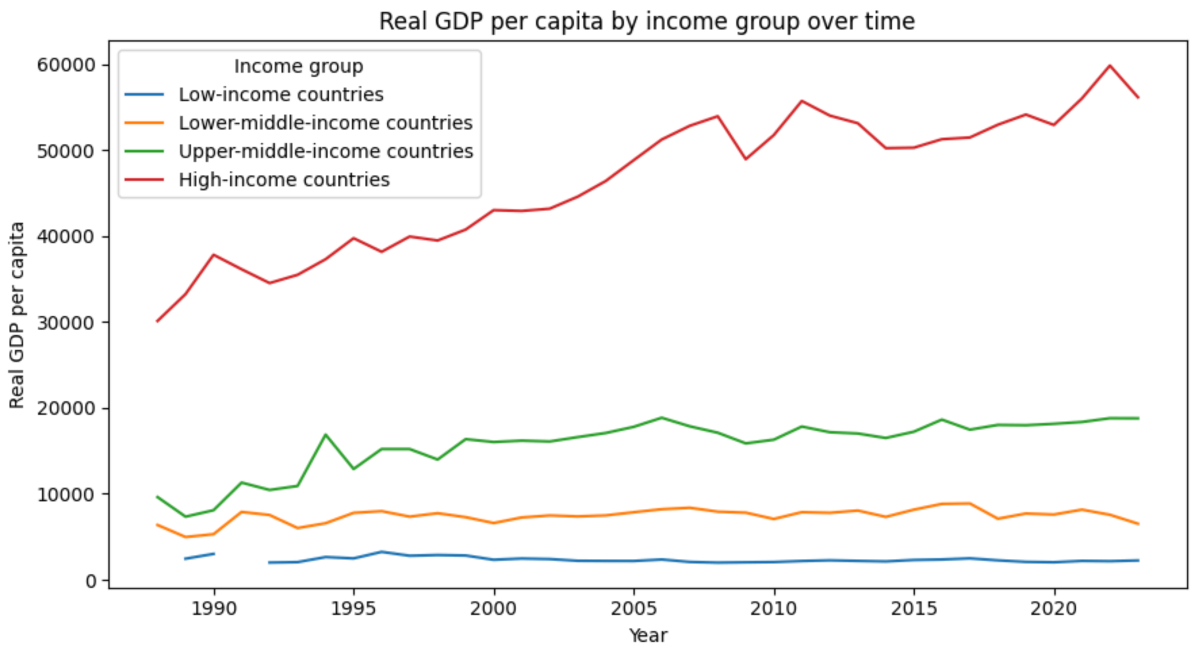
Figure 2

Line plot showing the trend of trade as percent of GDP by income group in the period 1988-2023

During this period of liberalisation, Figure 3 shows how the countries from different income groups see different rates of economic growth. High-income countries and upper-middle-income countries experience a sizeable real GDP per capita annual growth throughout, but low- and lower-middle-income countries stay largely stagnant throughout.

Figure 3

Line plot showing trend of real GDP per capita by income group in the period 1988-2023



Analysis

This analysis aims to replicate the model of Sachs & Warner (1995) and Wacziarg & Welch (2008), while replacing their openness dummy variable with lagged tariff rates. Income group classification dummies were also included to control for differences in growth rates across income levels not captured by the other control variables. This analysis sets out to use the same regression model by using the same control variables, or substituting with suitable replacements whenever data is not available. Including control variables ensures that the estimated effect of tariff rates on economic growth is not confounded by standard growth determinants.

The hypothesis for this analysis are the same as Sachs & Warner and Wacziarg & Welch's hypotheses, adapted to fit the model of the liberalisation dummy being replaced by a lagged tariff variable.
H0: Tariff rates have no impact on real GDP per capita growth, controlling for other factors.
HA: Tariff rates have a significant impact on real GDP per capita growth, controlling for other factors.

Following the specification of Sachs & Warner and Wacziarg & Welch's models with the changes displayed in Table 1, the following model is run:

$$log\_gdppc\_growth_{it} = \alpha + \beta_1 tariff_{it-1} + \beta_2 X_{it} + \varepsilon_{it},$$

where X contains time-varying control variables shown in Table 1.

Table 1 reports the regression model used in Sachs & Warner (1995) and Wacziarg & Welch's (2008) regression analysis, and the results of the regression of the replicated model, with lagged tariffs to replace the openness dummy variable. The first column states the names of variables used in their models, while the second column shows whether these variables were replicated exactly in the current analysis, were substituted by proxy variables, or were omitted. Notably, variables indicating political instability included in the original model were not included in this analysis due to data availability problems. The third column shows estimated coefficients.

Table 1

Replication of Sachs & Warner and Wacziarg & Welch Regression Model

Independent Variables		Dependent Variable
Sachs & Warner (1995), Wacziarg & Welch (2008)	Tariff Analysis	Log real GDP per capita growth (t)
Sachs-Warner openness dummy variable Openness status based on liberalization dates (t)	Tarrif rate (t-1)	-0.0355
Secondary-school enrollment rate (t) Primary-school enrollment rate (t)	Human capital	0.2943
Government Consumption to GDP ratio (t, t + X)	Share of government consumption at current PPPs	-9.1179*
Deviation of the price level of investment (t)	Price level of capital formation	-0.4148**
Gross domestic investment/ real GDP (t, t + X)	Share of gross capital formation at current PPPs	13.5843**
Population density (t-10)	Population growth rate	-54.9447**
Number of revolutions per year (t, t + X)	Not included	
Number of assassinations per capita per year (t, t + X)	Not included	
Extreme political repression	Not included	
Not included	Dummy for low-income country	3.2985*
Not included	Dummy for lower-middle-income country	2.3354**
Not included	Dummy for upper-middle-income country	1.8782**
Intercept		0.1615
Adjusted R2		0.051
Number of observations		2442

Note: The third column reports the coefficients of the regression analysis.
* $p < 0.05$
** $p < 0.001$
Sachs & Warner used import-weighted tariff data to construct their openness dummy variable, whereas Wacziarg & Welch used unweighted tariff data due to data availability problems. This analysis uses import-weighted tariff data.

Source: Sachs & Warner and Wacziarg & Welch's regression model.

The regression results indicate that lagged tariffs do not significantly impact real GDP per capita growth, thus failing to reject the null hypothesis. Several control variables are highly significant. Income group indicators are also significant, indicating that there are differences in growth rates across income levels.

Methodology

The methodology used for the analysis consists of data cleaning, data exploration, and running a regression model predicting log real GDP per capita growth. Data was analysed in Python.

The tariff dataset was processed to obtain a trade-weighted average tariff rate, representing the average rate of duty per imported value unit. Compared to a simple average, the most commonly used indicator which weighs all products equally, the weighted tariff is weighted by the value of imports and reflects real trade flows. A remark is that the import-weighted tariff tends to be lower than the simple average tariff, since high tariffs in product categories tend to reduce imports of those goods, lowering their weight in the calculation. Weighted tariffs were chosen to represent Sachs & Warner's analysis, as they use import-weighted tariff rates in order to calculate their openness dummy variable.

The dataset contains country-year tariff data for four product categories - raw materials, intermediate goods, consumer goods, and capital goods. For each category, the dataset reports a trade-weighted average ad valorem tariff and the corresponding import value in 1000 USD. An import-weighted average tariff is constructed for each country and year in order to obtain a single aggregated value to use for country-year tariff rates. The import-weighted tariff is computed by averaging the category-level tariff rates using import values as weights. This is done to ensure that categories that account for a larger share of a country's imports receive a greater weight in the final variable. An alternative approach would have been to take a simple average of the import-weighted tariff rates across the four product categories, however, this would give equal weight to categories regardless of their import shares, leading to some categories being overrepresented and others underrepresented. Instead, this method ensures weighting first within each product category, and then across all four product categories, accounting for differences in category imports, and resulting in a value that reflects the tariff rate faced by the average imported good in that country and year.

Real GDP data was used from the Penn World Table, version 11.0, released in October 2025, which is a large dataset providing macroeconomic time series data for many countries.

Data

Tariff data was downloaded from the World Bank's Integrated Trade Solution (WITS), using their United Nations Conference on Trade and Development (UNCTAD) Trade Analysis Information

System (TRAINS) database. The dataset contains information on MFN tariff rates from the period 1988-2023.
Trade data was sourced from World Bank, downloaded from Our World in Data. It measures exports and imports of goods and services as percent of GDP.
Economic data was used from Penn World Table (PWT). This is the same dataset used by Sachs & Warner (1995) and Wacziarg & Welch (2008), but it is the newest release of the data - version 11, published on October 7, 2025, instead of version 5 and version 6, respectively.
Income group classification data is from World Bank's income classification dataset, which defines annual income classifications based on countries' gross national income per capita.

Limitations

One limitation comes from using the WITS database for tariffs. According to Teti (2024), WITS data suffers from positive selection bias: while there are much higher trade volumes present in reality, WITS only reports a small subset of them, because WITS includes only importer-exporter product combinations with positive trade flows (Teti, 2024).

Missing Data Handling of Tariff Data

Missing tariff data posed a significant challenge to the analysis, requiring many decisions regarding its treatment. To illustrate this, the raw WITS dataset contains tariff data for the period 1988-2023; however, 87.94% of all non-missing observations are concentrated in the period 2000-2023. Reliable tariff data before 2000 is limited for the scale of the analysis pursued in this paper, so this constitutes a limitation in this research.

To understand how to handle the missing data, firstly, the nature of the missing data must be understood. The missing data present in this dataset is of the type where there is a ‘real’ value, which happens to be missing, as opposed to values that are not defined. Each country has had an imposed tariff rate on trade for each year; however, a report of that rate is missing in some years. Therefore, statistical methods to handle these missing real values were evaluated and used to preprocess the dataset.

Secondly, the way in which the data is missing must be acknowledged. This could be because of three reasons: the data is missing completely at random, missing at random, or not missing at random. For a variable to be defined as missing at random, the probability that its observation is missing must not be related to the value of that variable, given all other observed variables are controlled (Allison, 2010). Accordingly, tariff data would be missing at random if and only if the probability that a tariff rate is missing in a given year does not depend on the unobserved tariff rate, once all other variables in the model are controlled for. This assumption cannot be tested, but rather, it can be theoretically justified.

Using an imputation-based method to fill in missing values in the dataset was considered. Specifically, in applying full information maximum likelihood in order to estimate the parameters of the distribution given the observable data, and use those parameters to impute the missing values. The maximum likelihood approach to imputation missing data is a method commonly used in many linear structural equation models and longitudinal studies for handling missing data (Allison, 2010). However, a key assumption of maximum likelihood is that data is missing at random (Allison, 2010). Tariff rates reflect changes in trade policy, rather than random independent observations in time following a normal distribution, so there is no theoretical reason to assume normality. Further, tariff rates are reported by countries, and countries can fail to report tariffs consistently for each year, which means that missing tariff rates would also be influenced by factors not related to randomness. In the

WITS dataset used in this analysis for supplying tariff data (Figure 5 in Appendix), it can be observed that the missing data appear not to be distributed at random. There is a substantial amount of missing tariff data in the period 1988-1995, data is often missing in continuous blocks, rather than isolated years, and some countries have more missing data than others. Therefore, an argument that data in the tariff dataset is missing at random cannot be justified, hence violating the missing at random assumption that is required for an imputation-based method like maximum likelihood to be unbiased.

In order to proceed with the analysis, univariate partial listwise deletion was used during preprocessing to restrict the sample to only countries with a substantial amount of data. The dataset was filtered to keep countries with tariff data available for at least 21 years over the period 1988-2023, corresponding to a minimum data coverage of 58.3%. This was implemented through listwise deletion – removing countries with missing tariff data for more than 15 years during the time period, and keeping countries with data availability in 21 years or more of the dataset.

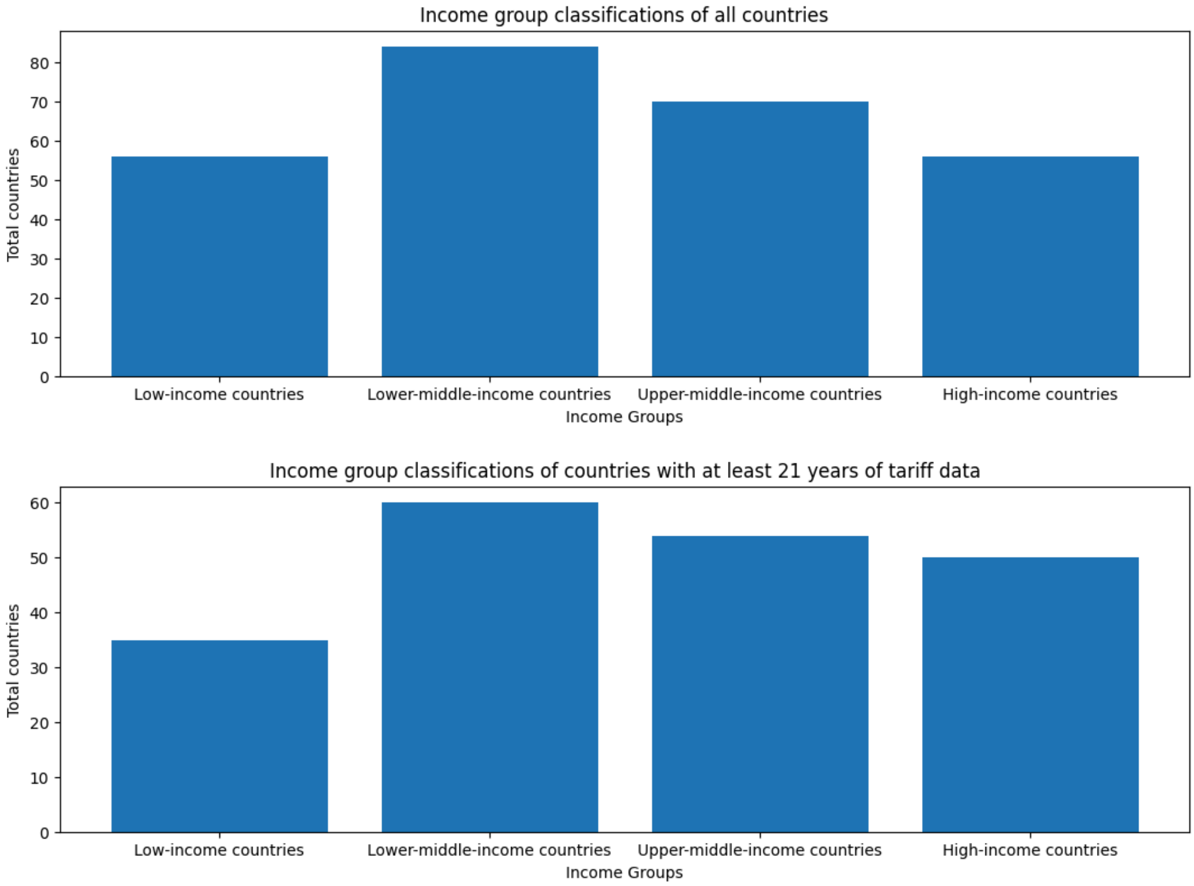
Listwise deletion was chosen for preprocessing because it is a common method for handling missing data, as it is straightforward and can be done with any statistical method, and results in estimated standard errors that estimate the true standard errors accurately (Allison 2010). For the tariff dataset in particular, listwise deletion is considered a good method for handling missing data of predictor variables because observations are not missing at random. Using listwise deletion will not make coefficient estimates biased, given that the probability of missing data of an independent variable is not related to the dependent variable (Allison, 2010). The only point of concern would then be a decrease in statistical power due to a smaller sample size; however, the total countries after applying listwise deletion is 109, compared to 144 prior, which still has the potential to yield a result with sufficient power.

The cut-off point of 21 years was chosen to remove countries with very low data availability that would realistically not be meaningful to include in the analysis, while keeping a balance of representation of countries from all income groups. This is important as the percentage of minimum tariff data availability increases, the ‘low-income countries’ group decreases faster in proportion to the other groups (Figure 7 in Appendix). This is because less developed countries had less data availability, leading them to lose representation in the dataset. Because the analysis hinges on a representative sample of countries from all income groups, a percentage of 58.3% was picked to best preserve the distribution of income groups from the unfiltered dataset, while still ensuring that countries with very little data were filtered out.

Additionally, Figure 4 shows that filtering the dataset for countries with at least 21 years of data did not meaningfully change the distribution of income groups featured in the sample. There is a very small change in the proportion of low-income to high-income countries; however, the dataset still sufficiently represents countries from all income groups.

Figure 4

Bar plots visualising the difference before and after filtering for missing data



AI Use Statement

Generative AI was used to generate Python code and assist in code debugging for data preprocessing, data visualisation, and data analysis. Generative AI was not used in the research process or writing process.

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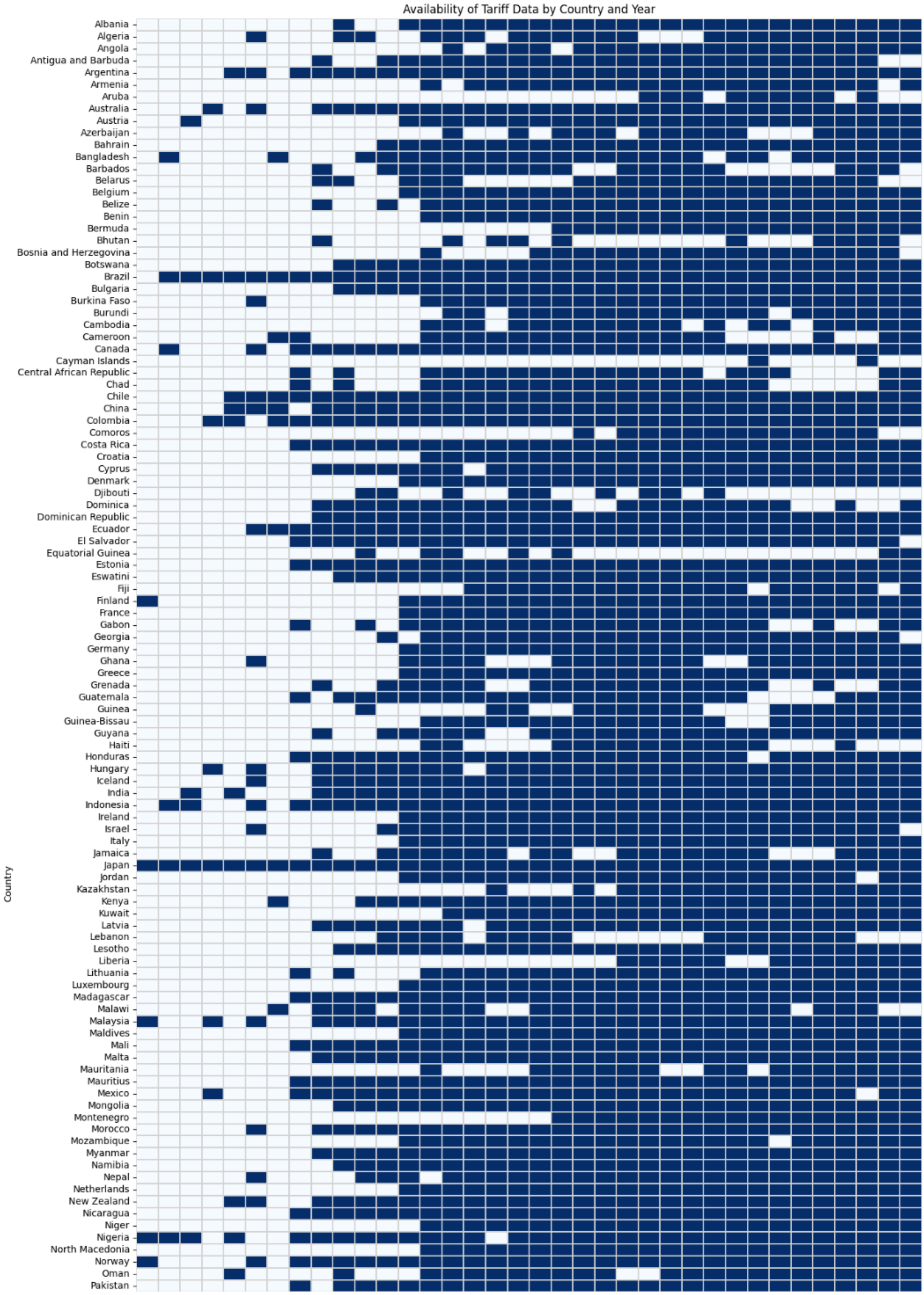
Appendix

Figure 5

Availability of tariff data by country and year (unfiltered, n=144)



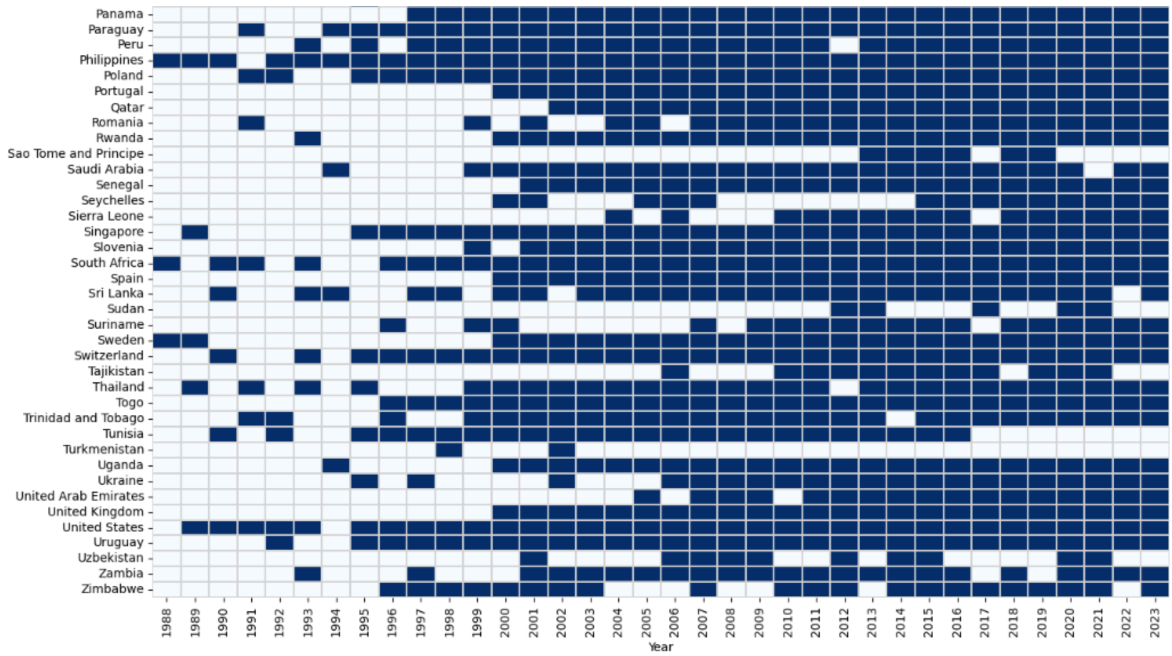
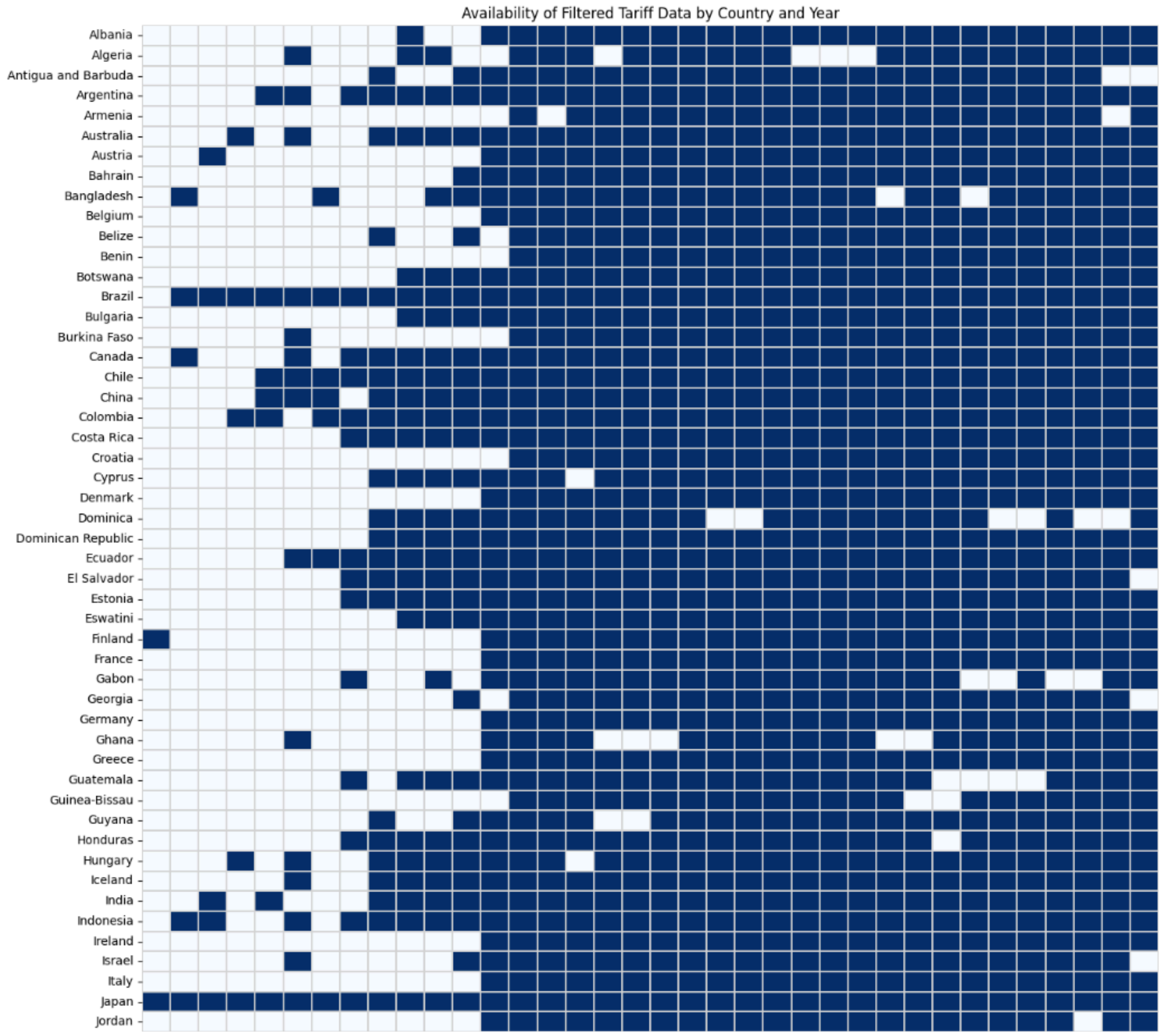
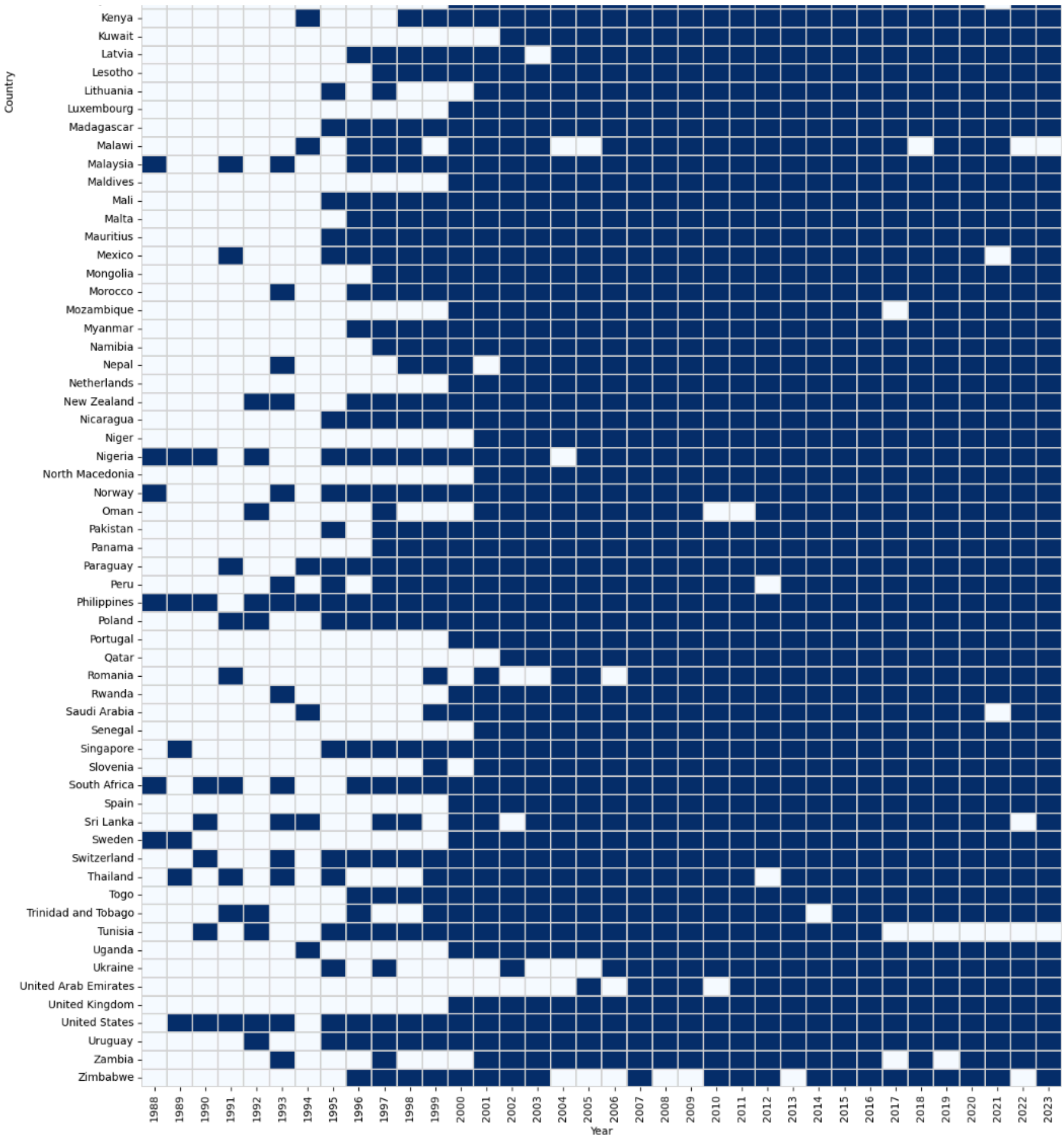


Figure 6

Filtered countries for 21 years or more of tariff data availability




Country
Kenya
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Mexico
Mongolia
Morocco
Mozambique
Myanmar
Namibia
Nepal
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New Zealand
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Rwanda
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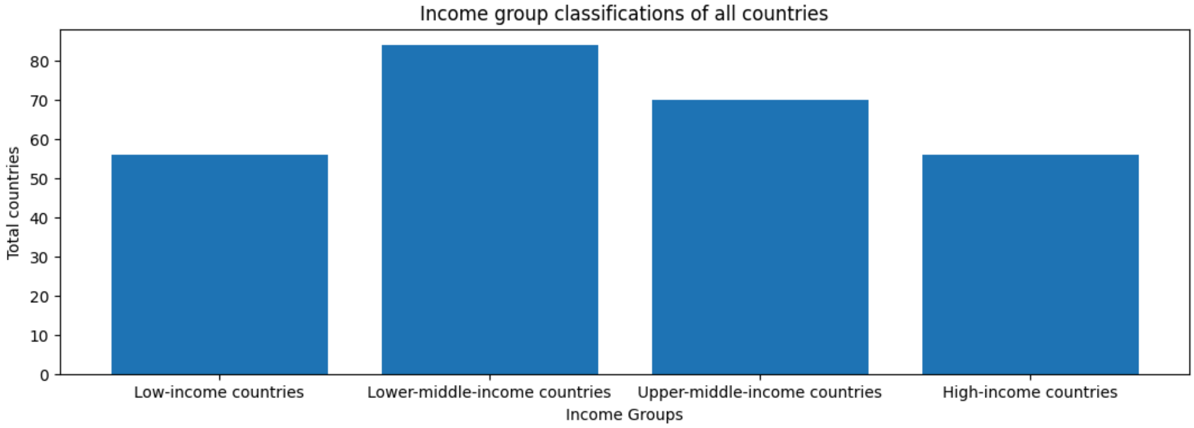
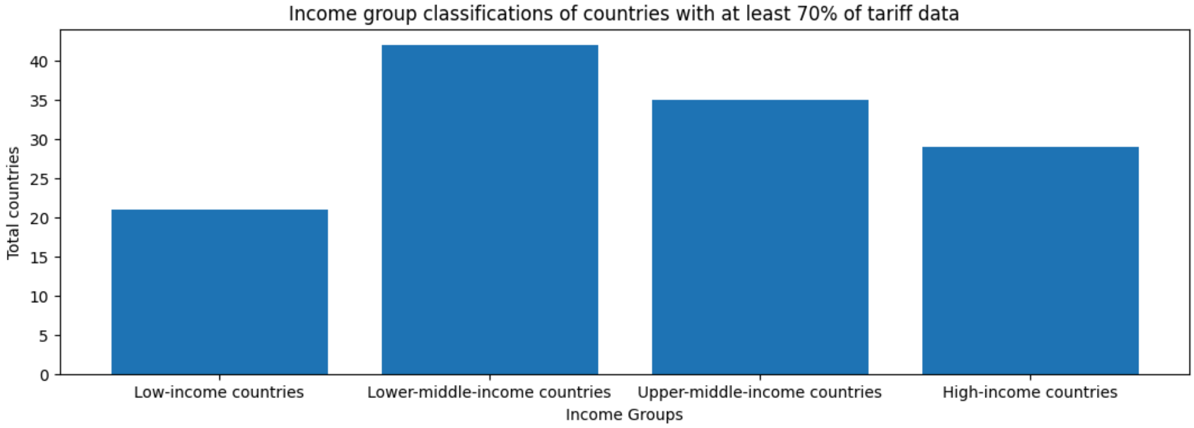
Figure 7

Income group distributions

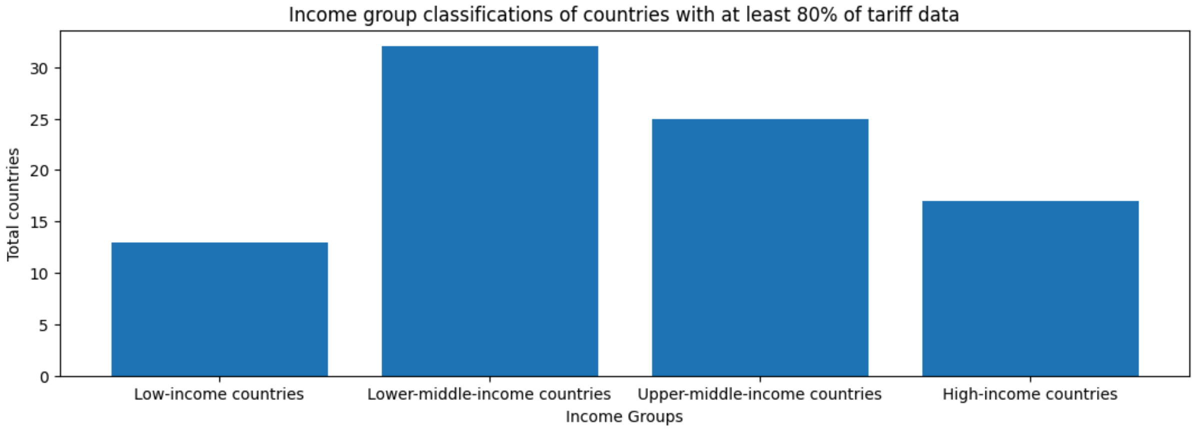
All countries (n = 144, min_years = 0)



Countries with at least 70% of tariff data availability (n = 68, min_years = 25)



Countries with at least 80% of tariff data availability (n = 47, min_years = 28)



Countries with at least 90% of tariff data availability (n = 9, min_years = 32)

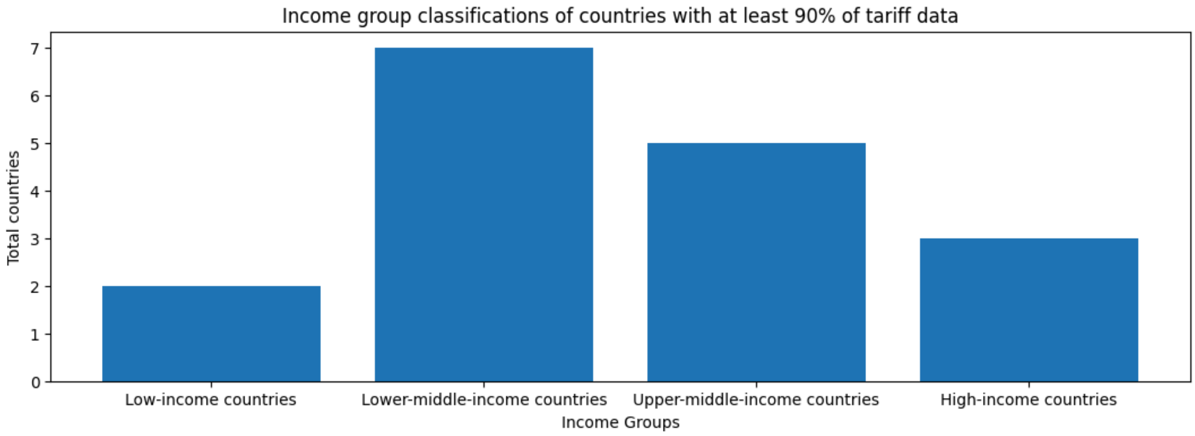


Figure 8

Heatmap showing income group categories of countries

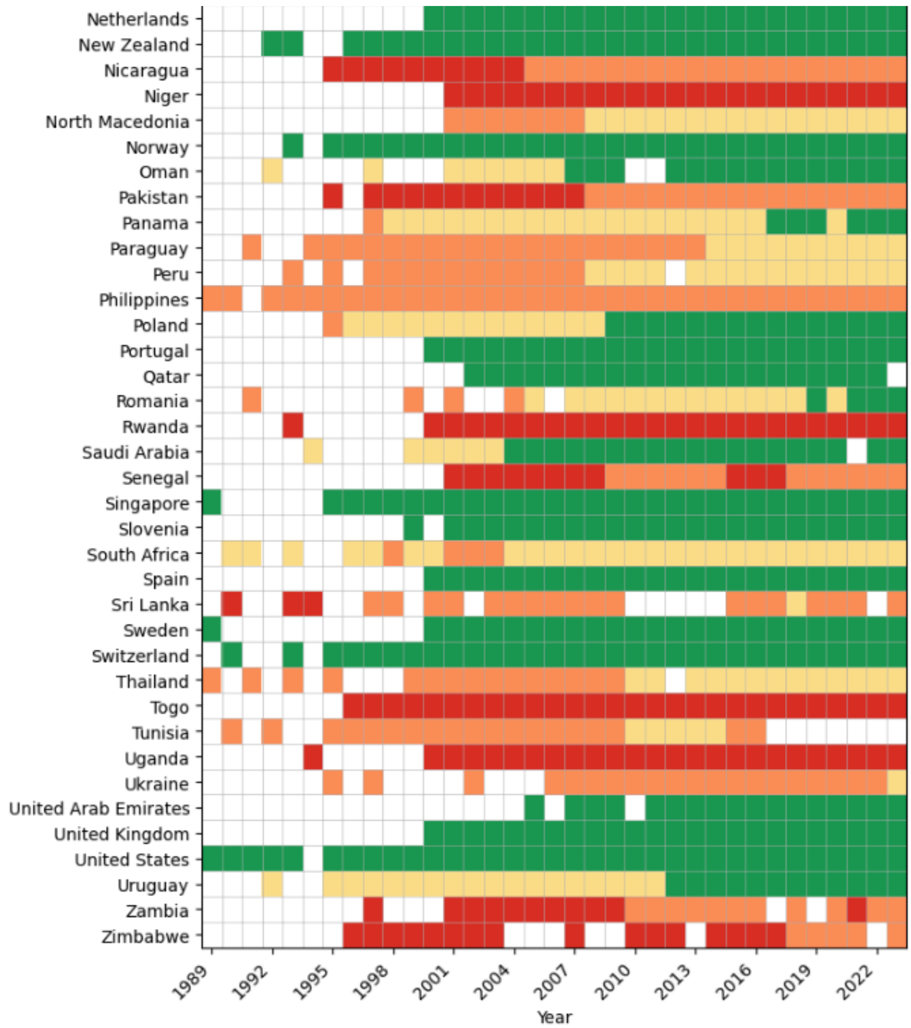


Table 2

Regression model output

```
                            OLS Regression Results
==============================================================================
Dep. Variable:        log_gdppc_growth   R-squared:                       0.055
Model:                             OLS   Adj. R-squared:                  0.051
Method:                  Least Squares   F-statistic:                     72.78
Date:                 Sun, 14 Dec 2025   Prob (F-statistic):           6.33e-39
Time:                         23:01:12   Log-Likelihood:                -7830.4
No. Observations:                 2442   AIC:                         1.568e+04
Df Residuals:                     2432   BIC:                         1.574e+04
Df Model:                            9
Covariance Type:               cluster
==============================================================================================================
                                                    coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------------------
Intercept                                         0.1615      1.701      0.095      0.924      -3.173       3.496
C(income_group)[T.Low-income countries]           3.2985      1.035      3.187      0.001       1.270       5.327
C(income_group)[T.Lower-middle-income countries]  2.3354      0.580      4.029      0.000       1.199       3.472
C(income_group)[T.Upper-middle-income countries]  1.8782      0.401      4.682      0.000       1.092       2.664
tariff_lag1                                      -0.0355      0.028     -1.273      0.203      -0.090       0.019
hc                                                0.2943      0.405      0.727      0.467      -0.499       1.087
csh_g                                            -9.1179      2.957     -3.084      0.002     -14.914      -3.322
pl_i                                             -0.4148      0.029    -14.257      0.000      -0.472      -0.358
csh_i                                            13.5843      2.604      5.216      0.000       8.480      18.688
pop_growth                                      -54.9447     13.258     -4.144      0.000     -80.930     -28.959
==============================================================================
Omnibus:                      549.966   Durbin-Watson:                   1.828
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            13850.680
Skew:                          -0.451   Prob(JB):                         0.00
Kurtosis:                      14.632   Cond. No.                         791.
==============================================================================

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
```